MCO Wave Parameter Library

ADDENDUM Rev D — FAA/IAC/Kershner/Mountain Flying Enhancements · 2026-06-07

Seven additions from literature review of: FAA Airplane Flying Handbook FAA-H-8083-3C (Chapters 4 and 10), FAA Mountain Flying FAA-P-8740-60, Kershner Advanced Pilot's Flight Manual, IAC Contest Rules, and STOL/bush flying literature. Supersedes no prior addenda — additive to Rev A, B, C v2.

Summary of Additions

#	Item	Source	Wave library symbol
1	Approach frequency ramp-down rate limit	FAA AFH Ch.4 MOSAIC Addendum — high-drag low-mass airplanes	APPROACH_FREQ_RAMP_DOWN_MAX_HZ_S = 0.20
2	Load-factor-adjusted stall speed	FAA AFH Ch.10 — steep turns; stall speed ∝ √load factor	stall_speed_banked(), stall_margin_banked_ms()
3	Stall recovery: accelerate first	FAA AFH Ch.4 — backside of power curve	stall_recovery_freq(), STALL_RECOVERY_FREQ_BOOST_HZ
4	Energy height tracking ES = h + V²/2g	FAA AFH Ch.4 — energy height map and trajectories	energy_height_m(), energy_height_rate_ms()
5	CHANDELLE mode — max-performance climb with	FAA AFH Ch.10; Kershner timing rules	chandelle_freq(), chandelle_bank(), CHANDELLE_*
6	SPIRAL DESCENT mode — controlled altitude loss	FAA AFH Ch.10 — steep spiral; SAR operational need	spiral_bank(), SPIRAL_*
7	MOUNTAIN MODE — terrain proximity envelope	FAA Mountain Flying FAA-P-8740-60; AOPA mountain rules	mountain_stall_floor_ms(), mountain_freq_bias(), mountain_mod

1. Approach Frequency Ramp-Down Rate Limit

Source: FAA AFH Ch.4 MOSAIC Addendum (October 2025)

The 2025 addendum to the FAA Airplane Flying Handbook adds a specific section on high-drag low-mass airplanes, noting that they decelerate rapidly when power is reduced. The MCO-1D (3.45 kg, large deployed wing area) is precisely this type. A sudden reduction in flap frequency is equivalent to a sudden throttle cut — it will cause rapid deceleration and potential loss of roll/yaw control on approach. The fix is simple: limit how fast the frequency can be reduced when descending.

Parameter / Function	Value	Purpose
`APPROACH_FREQ_RAMP_DOWN_MAX_HZ_S`	0.20	Hz/s — max frequency reduction rate on approach
`rate_limited_freq(current, target, dt, descending)`	function	Returns rate-limited frequency; cap only applies when descending=True

Validation test: T-RD1: Fly approach at 1.5Hz. Rapidly command 0.4Hz. Verify actual freq reduces at ≤ 0.20 Hz/s. Compare to unrestricted: aircraft should maintain airspeed more smoothly.

2. Load-Factor-Adjusted Stall Speed

Source: FAA AFH Chapter 10 — Steep Turns

In a banked turn, the wing must generate more lift to provide both vertical lift (to maintain altitude) and horizontal force (centripetal). The load factor equals 1/cos(bank). Stall speed increases as the square root of load factor. At 45° bank: Vs_turn = Vs × √(1/cos45°) = 1.19 × Vs. The current wave library stall protection uses only straight-flight Vs. In any banked turn the stall margin is actually smaller than indicated.

Parameter / Function	Value	Purpose
`stall_speed_banked(Vs_ms, bank_deg)`	function	Returns stall speed corrected for current bank angle
`stall_margin_banked_ms(V_ms, Vs_ms, bank_deg)`	function	Returns actual stall margin in a turn — may be negative when straight-flight margin looks positive
`performance_model: stall_speed_banked_ms`	added	Now included in every predict() return dict
`performance_model: stall_margin_banked_ms`	added	Alert fires when < 1.0 m/s AND bank > 20°

Validation test: T-LF1: Fly at 45° bank near stall. Verify stall_margin_banked_ms is 19% smaller than stall_margin_ms. Alert should fire at lower speed than straight-flight warning.

3. Stall Recovery — Accelerate First

Source: FAA AFH Chapter 4 — Backside of the Power Curve

On the backside of the power required curve (low speed, high AoA), increasing thrust alone does not recover — the aircraft is in a region of reversed command where reducing AoA (accelerating) is required first. For the MCO this means: when stall protection triggers, the first action must be a frequency boost to accelerate, not a climb attempt. Attempting to climb from the backside wastes energy and may deepen the stall.

Parameter / Function	Value	Purpose
`STALL_RECOVERY_FREQ_BOOST_HZ`	0.30	Hz added during 2-second recovery phase
`STALL_RECOVERY_BOOST_DURATION`	2.0	seconds — duration of acceleration phase
`stall_recovery_freq(base_hz, t_since_trigger)`	function	Returns commanded frequency: boosted for first 2s, then returns to base

Validation test: T-SR1: Approach stall from below. Verify auto-stall recovery commands +0.3 Hz for 2 seconds before returning to cruise frequency. Compare recovery altitude to simple frequency-hold recovery.

4. Energy Height Tracking ES = h + V²/2g

Source: FAA AFH Chapter 4 — Energy Height Map and Trajectories

The FAA Ch.4 defines energy height ES = h + V²/2g — the altitude equivalent of the aircraft's total specific energy. When ES falls rapidly the aircraft is spending stored energy (altitude + kinetic) faster than thrust replenishes it. This complements the Boyd Ps (which measures the rate of energy gain/loss from thrust/drag) with a direct measure of total energy state. Alert threshold: ES falling faster than 1.0 m/s sustained.

Parameter / Function	Value	Purpose
`energy_height_m(h_agl_m, V_ms)`	function	Returns ES = h + V²/2g in metres
`energy_height_rate_ms(es_now, es_prev, dt)`	function	Returns dES/dt — negative when energy depleting
`ENERGY_HEIGHT_ALERT_RATE_MS`	1.0	m/s — alert threshold
`performance_model: energy_height_m`	added	Logged each loop; included in delta report
`performance_model: energy_height_rate_ms`	added	Alert fires if ES falling > 1.0 m/s outside stoop

Validation test: T-EH1: Fly level at constant throttle. Verify ES ≈ constant. Then reduce throttle — verify ES drops and alert fires within 1s. Compare ES_rate with Ps reading: both should be negative simultaneously.

5. CHANDELLE Mode — Maximum Performance Climbing Turn

Source: FAA AFH Chapter 10; Kershner Advanced Pilot's Flight Manual

The chandelle is the most efficient maneuver for gaining altitude while changing direction 180°. It converts kinetic energy (airspeed) to potential energy (altitude) while simultaneously turning. Kershner's rule: maximum pitch (frequency) at exactly 90° of heading change; wings level at exactly 180°. For SAR operations near terrain this is the correct escape maneuver — more altitude gained per distance traveled than a straight climb, while also reversing course away from terrain.

Parameter / Function	Value	Purpose
`CHANDELLE_BANK_DEG`	30.0°	Bank angle for the climbing turn
`CHANDELLE_ENTRY_V_FACTOR`	1.40	Entry speed = Vs × 1.40 minimum
`CHANDELLE_FREQ_RAMP_START`	0.80 Hz	Frequency at entry
`CHANDELLE_FREQ_RAMP_END`	2.20 Hz	Frequency at 90° heading point
`CHANDELLE_SWEEP_DEG`	60°	All wings deployed for maximum lift
`chandelle_freq(heading_change_deg)`	function	Returns frequency as function of heading change (0–180°)
`chandelle_bank(heading_change_deg)`	function	Returns bank angle: constant 30° for first 90°, linear rollout through second 90°

Validation test: T-CH1: Enter from Vs×1.4, 30° bank. Verify max frequency at exactly 90° heading change. Verify wings level at exactly 180°. Measure altitude gained vs straight climb at same entry speed and duration.

6. SPIRAL DESCENT Mode — Controlled Altitude Loss Over a Point

Source: FAA AFH Chapter 10 — Steep Spiral; SAR operational requirement

The steep spiral allows rapid altitude loss while maintaining visual contact with a reference point below — the correct technique for descending to a rescue site in confined terrain. Bank angle is modulated for wind (steeper downwind, shallower upwind) to maintain a constant-radius circle over the target. Minimum exit altitude 50m AGL. The MCO-1D's existing stoop is a straight-line dive — the spiral is more appropriate for SAR operations where the rescue site must remain in view.

Parameter / Function	Value	Purpose
`SPIRAL_BANK_DEG_NOMINAL`	45°	Bank in calm air
`SPIRAL_BANK_DEG_MAX`	55°	Maximum (downwind heading)
`SPIRAL_BANK_DEG_MIN`	30°	Minimum (upwind heading)
`SPIRAL_FREQ_HZ`	1.0 Hz	Moderate thrust for controlled descent
`SPIRAL_MIN_EXIT_AGL_M`	50 m	Minimum altitude to begin pullout
`SPIRAL_DESCENT_RATE_MS`	1.5 m/s	Target descent rate
`spiral_bank(heading, wind_from, wind_speed)`	function	Returns wind-corrected bank angle for constant-radius turn

Validation test: T-SP1: Enter spiral over a ground reference point at 100m AGL. Verify constant radius maintained over 3 turns (GPS track). Verify bank steeper when heading downwind. Verify auto-exit at 50m AGL.

7. MOUNTAIN MODE — Terrain Proximity Envelope Tightening

Source: FAA Mountain Flying FAA-P-8740-60; AOPA Mountain Flying; FAA AIM §7-6-7

Mountain flying demands tighter margins than flatland operations. The FAA recommends crossing ridges at +1,000ft (scaled to +300m for MCO-1D), maintaining wider pattern turns (increased TAS at altitude), and never descending into confined terrain without a known escape route. Mountain mode activates when GPS terrain data shows high ground within 500m laterally. It raises the stall floor, biases frequency upward to maintain positive Ps against terrain, and blocks stoop modes (which require altitude the terrain may not provide).

Parameter / Function	Value	Purpose
`MOUNTAIN_MODE_TERRAIN_RADIUS_M`	500 m	Lateral range to trigger mountain mode
`MOUNTAIN_MODE_FREQ_BIAS_HZ`	+0.20 Hz	Frequency addition to maintain Ps margin near terrain
`MOUNTAIN_MODE_STALL_FACTOR`	1.25×	Stall floor = Vs × 1.25 in mountain mode (vs normal 1.10×)
`MOUNTAIN_MODE_MIN_RIDGE_CLEAR_M`	300 m	Minimum AGL clearance over terrain (FAA 1,000ft rule scaled)
`mountain_stall_floor_ms(Vs, mountain_mode)`	function	Returns appropriate stall floor for mode
`mountain_freq_bias(freq, mountain_mode)`	function	Returns frequency with terrain bias applied
`mountain_mode_blocks_stoop(mountain_mode)`	function	Returns True if stoop should be blocked

Validation test: T-MT1: Activate mountain mode (set px.mountain_mode=True in sim). Verify stall floor increases from 1.10× to 1.25× Vs. Verify frequency bias +0.20 Hz active. Verify stoop command blocked with status message.

Open Items Added by Rev D

#	Item	Priority
O-D1	T-RD1: Approach ramp-down rate test	Medium
O-D2	T-LF1: Load-factor stall in turns test	High — safety
O-D3	T-SR1: Stall recovery sequence test	High — safety
O-D4	T-EH1: Energy height tracking validation	Medium
O-D5	T-CH1: Chandelle mode flight test	Medium
O-D6	T-SP1: Spiral descent over ground point	Medium

#	Item	Priority
O-D7	T-MT1: Mountain mode activation and limits	High — SAR critical
O-D8	GPS terrain proximity detection — FCC integration needed	High — mountain mode trigger depends on th